

7. Test Environment Setup

7.1 Environment Setup

Introduction

The Environment Setup defines the configuration of all required tools, software, and dependencies needed to perform security testing and automation testing of the OWASP Juice Shop application. It ensures that the testing environment is stable, consistent, and capable of executing Selenium WebDriver and TestNG-based automation scripts without failures.

A properly configured environment is essential for achieving reliable test execution results, minimizing setup-related issues, and ensuring smooth integration between the application and automation framework.

Objective

- To configure a stable testing environment
 - To install required tools for automation testing
 - To ensure compatibility between browser, driver, and Selenium
 - To support execution of security test cases
 - To reduce environment-related failures during test execution
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Software Requirements

- Operating System: Windows 10 / Windows 11
 - Java JDK 8 or higher
 - Eclipse IDE
 - Maven
 - Selenium WebDriver
 - TestNG Framework
 - Google Chrome Browser
 - ChromeDriver (matching version)
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Configuration Steps

- Java installation and environment variable setup (JAVA_HOME)
- Eclipse IDE installation and workspace setup

- Maven project creation and dependency configuration
 - Selenium WebDriver dependency setup in pom.xml
 - TestNG plugin installation in Eclipse
 - ChromeDriver configuration for browser automation
 - Framework structure setup (POM architecture)
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Outcome

A fully functional automation testing environment was created, capable of executing security test scripts for OWASP Juice Shop efficiently.